

Question number	Answer	Additional guidance	Mark
7(a)(i)	3500 nm / 3.5×10^3 nm / 3.5 μm (1)	ACCEPT {3 / 3.3} {× / times} DO NOT ACCEPT correct value with wrong unit 3 / 3.3 with units 3 500 or 3.5 without units	(1)

Question number	Answer	Mark																			
7(a)(ii)	<table><tr><th rowspan="2">Statement about cells</th><th colspan="4">Type of microorganism</th></tr><tr><th>both <i>S. cerevisiae</i> and <i>S. aureus</i></th><th><i>S. cerevisiae</i> only</th><th><i>S. aureus</i> only</th><th>neither <i>S. cerevisiae</i> nor <i>S. aureus</i></th></tr><tr><td>Contain both DNA and RNA</td><td>☒</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>Have membranes around the cytoplasm and around the nucleus</td><td>☐</td><td>☒</td><td>☐</td><td>☐</td></tr></table>	Statement about cells	Type of microorganism				both <i>S. cerevisiae</i> and <i>S. aureus</i>	<i>S. cerevisiae</i> only	<i>S. aureus</i> only	neither <i>S. cerevisiae</i> nor <i>S. aureus</i>	Contain both DNA and RNA	☒	☐	☐	☐	Have membranes around the cytoplasm and around the nucleus	☐	☒	☐	☐	(2)
Statement about cells	Type of microorganism																				
	both <i>S. cerevisiae</i> and <i>S. aureus</i>	<i>S. cerevisiae</i> only	<i>S. aureus</i> only	neither <i>S. cerevisiae</i> nor <i>S. aureus</i>																	
Contain both DNA and RNA	☒	☐	☐	☐																	
Have membranes around the cytoplasm and around the nucleus	☐	☒	☐	☐																	

Question number	Answer	Additional guidance	Mark
7(b)(i)	• explanation of how the number of yeast cells are determined that includes dealing with yeast on the edges and matches their number if given (= 16 / 17 / 18) (1) • $(0.00625 \mu\text{l} \Rightarrow) 0.00000625 / 6.25 \times 10^{-6} (\text{cm}^3) (1)$ • (concentration $\Rightarrow$) 2 560 000 / 2 600 000 / 2.56×10^6 / 2.6×10^6 / 2 720 000 / 2 700 000 / 2.72×10^6 / 2.7×10^6 / 2 880 000 / 2 900 000 / 2.88×10^6 / 2.9×10^6 (cells per cm^3) (1)	e.g. 16 + 2 halves = 17 cells touching two sides are counted in IGNORE ref to living / dead / exclusion dyes ecf if mp2 is correct except for wrong order of magnitude Bald answer of 2 560 000 / 2 720 000 / 2 880 000 etc gets 2 max	(3)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An answer that includes the following points: • both axes labelled : log (living) {(number)cells / yeast} and time (1) • units for x axis : hours (1) • shape of curve drawn correctly (1) • all four phases labelled correctly (1)	ACCEPT $\ln$ / $\log_{10}$ ACCEPT minutes i.e. lag + {log / exp} + stationary + {death / decline} Log cells  Time / hrs	(4)

Question number	Answer	Additional guidance	Mark
7(b)(iii)	A description that includes the following points: • indication of where log values have come from (1) • during {log / exponential} phase (1) • determine t / subtract the two time values (1) • growth rate equation used to calculate the constant (1)	e.g $\log_{10}$ {concentration / number} read from a log graph calculated from actual numbers (from a graph) ACCEPT description e.g. at end of lag phase and beginning of stationary phase ACCEPT from a reasonable attempt at giving the formula ACCEPT from a reasonable attempt at giving the formula $k = \frac{\log N_t - \log N_0}{0.301t}$	(4)

